III. Write TRUE or FALSE:

1. Println prints an item in the next line.
2. Local variables also exist outside the function.
3. Overloading of methods even depends on return type.
4. A function cannot be defined without parameters.
5. A function body is enclosed within curly brackets.

IV. Answer the following:

1. Define a function. What do you mean by function prototype?
2. What are the two ways of invoking functions?
3. When a function returns the value, the entire function call can be assigned to a variable. Do you agree with the statement? [ICSE 2011]
4. When a function is invoked how many values can be returned from the function?
5. Debug the errors and rewrite in the following function prototype:
   (a) int sum(x,y);
   (b) float product(a,int y);
   (c) float operate(int x, float=3.4);
   (d) float sum(int x,y);
6. For the given function write down the main function which calls it:

```
int square(int a)
{
    return(a*a);
}
```

7. Mention the reason for using pass by reference operation. What happens when a function is passed by reference?
8. In what situation a function returns a value?
9. Differentiate pure and impure functions. [ICSE 2009]
10. Write a function which is used to swap the values of two memory locations.
    (a) by using third variable.
    (b) without using third variable.
11. Differentiate between call by value and call by reference.
12. What are the advantages of defining a method in a program? [ICSE 2010]
13. What do you mean by function overloading? In what way it is advantageous?
14. Define the following:
    (a) Pure Function [ICSE 2006]
    (b) Impure Function [ICSE 2005]
    (c) Access Specifier [ICSE 2006]
    (d) Parameter list
    (e) Recursive Function
    (f) Method Signature
15. Explain the function of a return statement in Java programming. [ICSE 2006]
16. Differentiate between Formal Parameter and Actual Parameter. [ICSE 2008]
17. What is the role of the keyword void in declaring functions? [ICSE 2007, 2011]
18. If a function contains several return statements, how many of them will be executed? [ICSE 2007]
19. Which OOP principle implements function overloading? [ICSE 2007]
20. How are the following data passed to a function?
    • Primitive types • Reference types [ICSE 2005]

V. Give the output in the following function definitions and also write what mathematical operations they carry out:

1.
```
void test1(int n)
{
for(int x=1; x<=n; x++)
if(n%x == 0)
      System.out.println(x);
}
```
if 12 is passed to n.

2.
```
void test 2(int a, int b)
{
while( a != b)
      {
         if ( a > b)
            a = a – b;
            else
            a = b – a;
      }
      System.out.println(a);
      }
```
if 4 and 17 are passed to the function.

3.
```
void test 3(char c)
{
      System.out.println( (int) c);
}
```
if 'm' is passed to c.

4.
```
void test 4(String x, String y)
{
      if(x.compareTo(y) > 0)
            System.out.println(x);
            else
            System.out.println(y);
}
```
if "AMIT" and "AMAN" are passed to the function.

VI. Java Programming on functions:

1. Write a program in Java to input 10 numbers. Use a function to find and print the cube of each number.
2. Write a Java program to input the sides of a triangle. Pass the sides to a function decide(int x,int y,int z) which checks and prints whether the triangle is equilateral, isosceles or scalene.
3. Write a program in Java to input day number. Use a function daysofweek(int dysno) to accept the day number from the main class. The function uses a switch-case statement to print the corresponding day of the week.
4. Write a Java program to input a number. Use a function int Armstrong(int n) to receive the number. The function returns 1 if the number is Armstrong otherwise 0.
5. Write a function fact(int n) to find the factorial of a number n. Include a main class to find the value of S where:

$$S = \frac{n!}{m!(n-m)!}$$

6. Write a Java program to accept a string. Pass it to a function freq(String x). The function finds and prints the frequency of each vowel.
7. Write a program in Java to accept a word. Pass it to a function magic(string x). The

function checks the string for the presence of consecutive letters. If two letters are consecutive in any position the function prints "It is a magic string" otherwise prints "It is not a magic string".

8. Write a program in Java to input a string using upper and lower case characters. Pass the string to a function lower(String x) to print only lower case letters.

9. Write a program in Java to accept 10 numbers in a Single Dimensional Array. Pass the array to a function swap(int a[]) to interchange the numbers in such a way that a[0] is interchanged with a[1], a[2] with a[3] and so on. Finally print all the numbers of the array.

10. Write a program in Java to accept 10 numbers in an array. Define a function int prime(int x) to check whether a number is prime or not. It returns 1 if the number is prime, 0 otherwise. Use the function to check and print all the prime numbers of the array.

11. Write a program using a function called area() to compute the area of a:
    (a) circle ($\pi * r^2$) where $\pi = 3.14$
    (b) square (side * side)
    (c) rectangle (length * breadth)

    Display the menu to output the area as per User's choice. [ICSE 2005]

12. Write a program using a method Palin( ), to check whether a string is a Palindrome or not. A Palindrome is a string that reads the same from the left to right and vice versa.
    Sample Input: MADAM, ARORA, ABBA etc. [ICSE 2007]

13. Write a program in Java using a method Discount( ), to calculate a single discount or a successive discount. Use overload methods Discount (int), Discount(int,int) and Discount(int,int,int) to calculate single discount and successive discount respectively. Calculate and display the amount to be paid by the customer after getting discounts on printed price of an article.

    Sample Input: Printed price: ₹ 12000

    Successive discounts = 10%, 8%
    = ₹ (12000 – 1200)
    = ₹ (10800 – 864)

    Amount to be paid = ₹ 9936

14. Write a class with the name Area using function overloading that computes the area of a parallelogram, a rhombus and a trapezium.

    Formula: Area of a parallelogram (pg) = b*h
    Area of a rhombus (rh) = ½*d1*d2 (where, d1 and d2 are the diagonals)
    Area of a trapezium (tr) = ½*(a + b)*h (where, a and b are the parallel sides, h = perpendicular distance between the parallel sides)

15. Write a class with the name Perimeter using function overloading that computes the perimeter of a square, a rectangle and a circle.

    Formula: Perimeter of a square = 4*s
    Perimeter of a rectangle = 2*(l +b)
    Perimeter of a circle = 2*π*r

16. Write a program in Java to accept a String from the user. Pass the String to a function Display (String str) which displays the consonants present in the String.

    Sample Input : computer
    Sample Output : c
    m
    p
    t
    r

17. Write a program in Java to accept a String from the user. Pass the String to a function Change (String str) which displays the first character of each word after changing the case (lower to upper and vice-versa).

Sample Input : Delhi public school
Sample Output : d
P
S

18. Write a program in Java to accept the name of an employee and the annual income. Pass the name and the annual income to a function Tax (String name, int income) which displays the name of the employee and the income tax as per the given tariff:

Annual Income	Income Tax
Up to ₹ 50,000	No tax
₹ 50,001 to ₹ 1,00,000	10% of the income exceeding ₹ 50,000
₹ 1,00,001 to ₹ 1,50,000	₹ 5,000 + 20% of the amount exceeding ₹ 1,00,000
₹ 1,50,001 and above	₹ 15,000 + 30% of the amount exceeding ₹ 1,50,000

19. Write a program in Java to accept a String from the user. Pass the String to a function First (String str) which displays the first character of each word.

Sample Input : Understanding Computer Applications
Sample Output : U
C
A

20. Design a class overloading a function display ( ) as follows:
    (i) void display (String str, int p) with one String argument and one integer argument. It displays all the uppercase characters if 'p' is 1 (one) otherwise, displays all the lowercase characters.
    (ii) void display (String str, char chr) with one String argument and one character argument. It displays all the vowels if chr is 'v' otherwise, displays all the alphabets.

21. Design a class overloading a function calculate ( ) as follows:
    (i) void calculate (int m, char ch) with one integer argument and one character argument. It checks whether the integer argument is divisible by 7 or not, if ch is 's' otherwise, it checks the last digit of the integer argument contains 7 or not.
    (ii) void calculate (int a, int b, char ch) with two integer arguments and one character argument. It displays the greater of integer arguments if ch is 'g' otherwise, displays the smaller of integer arguments.

22. Write a program using method name Glcm(int,int) to find the Lowest Common Multiples (LCM) of two numbers by GCD (Greatest Common Divisor) of the numbers.
GCD of two integers is calculated by continued division method. Divide the larger number by the smaller, the remainder then divides the previous divisor. The process is repeated till the remainder is zero. The divisor then results in the GCD.

$$\text{LCM} = \frac{\text{Product of two numbers}}{\text{GCD}}$$

Write the main method to display the LCM of two numbers.

23. Write a program to accept 10 numbers in a Single Dimensional Array. Pass the array to a function Search(int m[], int ns) to search the given number ns in the list of array elements. If the number is present, then display the message 'Number is present' otherwise, 'number is not present'.

24. Write a menu driven program using a method Number ( ) to perform the following tasks:
    1. Accept a number from the user and display in its **Binary Equivalents.**
       e.g.: Sample Input : $(21)_{10}$, Sample Output: $(10101)_2$
    2. Accept a binary number from the user and display in its **Decimal Equivalents.**
       e.g.: Sample Input : $(11101)_2$, Sample Output: $(29)_{10}$

25. Design a class to overload a function manip ( ) as follows:
    (i) void manip (String str, int p) with one String argument and one integer argument. It displays the characters of even positions of String, if p is an even number otherwise, displays the characters of odd positions.
    (ii) void manip (int a, char ch) with one integer argument and one character argument. It computes the square root of integer arguments if ch is 's' else cube root of the integers.

26. Design a class to overload a function manip ( ) as follows:
    (i) void compare (int, int) : to compare two integers values and print the greater of the two integers.
    (ii) void compare (char, char) : to compare the numeric value of two characters and print with the higher numeric value.
    (iii) void compare (String, String) : to compare the length of the two strings and print the longer of the two. [ICSE 2011]

27. Design a class to overload a function polygon( ) as follows:
    (i) void polygon(int n, char ch): with only integer and one character type argument to draw a filled square of side n using the character stored in ch.
    (ii) void polygon(int x, int y) : with two integer arguments that draws a filled rectangle of length x and breadth y, using the symbol '@'.
    (iii) void polygon( ) : with no argument that draws a filled rectangle shown below: [ICSE 2012]

    Example:
    (i) Input value of n=2, ch = 'O'
    Output: OO
    OO
    (ii) Input value of x = 2, y = 5
    Output: @@@@@
    @@@@@
    (iii) Output: *
    **
    ***

28. Design a class to overload a function series ( ) as follows:
    (i) double series (double n) with one double argument and returns the sum of the series.
    sum = $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \ldots\ldots\ldots\ldots\ldots + \frac{1}{n}$
    (ii) double series(double a, double n) with two double arguments and returns the sum of the series.
    sum = $\frac{1}{a^2} + \frac{4}{a^5} + \frac{7}{a^8} + \frac{10}{a^{11}} +$ ---------- to n terms [ICSE 2013]